Name
Date
Course

Beginning Algebra Final Test (Part I) — Practice Test 1

Instructions

i) As always, write a title, name, date, and course.
ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
iii) Draw a rectangle around your answers.
iv) Staple a copy of these test questions with your work and your final answers.

Solve.

① $5y^6 - 505y^4 = -500y^2$

$5y^6 - 505y^4 + 500y^2 = 0$

$5y^2(y^4 - 101y^2 + 100) = 0$

$5y^2(y^2 - 1)(y^2 - 100) = 0$

$5y^2(y-1)(y+1)(y-10)(y+10) = 0$

$5y^2 = 0$ or $y - 1 = 0$ or $y + 1 = 0$ or $y - 10 = 0$ or $y + 10 = 0$

$\boxed{y = 0}$ or $\boxed{y = 1}$ or $\boxed{y = -1}$ or $\boxed{y = 10}$ or $\boxed{y = -10}$

② $-3x^5 - 27x^3 = 0$

$-3x^3(x^2 + 9) = 0$

$-3x^3 = 0$ or $x^2 + 9 = 0$

$\boxed{x = 0}$ or $x^2 \neq -9$

③ $2b^3 = 16b^2 + 18b$

$2b^3 - 16b^2 - 18b = 0$

$2b(b^2 - 8b - 9) = 0$

$2b(b - 9)(b + 1) = 0$

$2b = 0$ or $b - 9 = 0$ or $b + 1 = 0$

$\boxed{b = 0}$ or $\boxed{b = 9}$ or $\boxed{b = -1}$

④ $x^3 + 8x^2 = x + 8$

$x^3 + 8x^2 - x - 8 = 0$

$x^2(x+8) - 1(x+8) = 0$

$(x+8)(x^2 - 1) = 0$

$(x+8)(x-1)(x+1) = 0$

$x + 8 = 0$ or $x - 1 = 0$ or $x + 1 = 0$

$\boxed{x = -8}$ or $\boxed{x = 1}$ or $\boxed{x = -1}$

Perform the indicated operations and simplify completely.

⑤ $$\frac{6x^2 - 5xy - 6y^2}{14x^2 - 21xy} \cdot \frac{7x^2y^5}{45x^3y - 20xy^3}$$

$$\frac{(2x-3y)(3x+2y)}{7x(2x-3y)} \cdot \frac{7x^2y^5}{5xy(9x^2 - 4y^2)}$$

$$\frac{(2x-3y)(3x+2y)}{7x(2x-3y)} \cdot \frac{7x^2y^5}{5xy(3x+2y)(3x-2y)}$$

$$\frac{x^2y^5}{5x^2y(3x-2y)}$$

$$\boxed{\frac{y^4}{5(3x-2y)}}$$

⑥ $\frac{1}{2b} - \frac{8b-3}{5b^2+b}$

$\frac{1}{2b} - \frac{8b-3}{b(5b+1)}$

$\frac{1(5b+1)}{2b(5b+1)} - \frac{2(8b-3)}{2b(5b+1)}$

$\frac{5b+1}{2b(5b+1)} - \frac{16b-6}{2b(5b+1)}$

$\frac{5b+1-16b+6}{2b(5b+1)}$

$\boxed{\frac{-11b+7}{2b(5b+1)}}$

⑦ $-\frac{2x-5}{x+7} + \frac{2}{x-3}$

$-\frac{(2x-5)(x-3)}{(x+7)(x-3)} + \frac{2(x+7)}{(x-3)(x+7)}$

$-\frac{2x^2-11x+15}{(x+7)(x-3)} + \frac{2x+14}{(x-3)(x+7)}$

$\frac{-2x^2+11x-15}{(x+7)(x-3)} + \frac{2x+14}{(x-3)(x+7)}$

$\boxed{\frac{-2x^2+13x-1}{(x+7)(x-3)}}$

⑧ $\frac{2x^5+24x^4+64x^3}{x^2-64} \div \frac{-9x^2}{80-10x}$

$\frac{2x^5+24x^4+64x^3}{x^2-64} \cdot \frac{80-10x}{-9x^2}$

$\frac{2x^3(x^2+12x+32)}{(x-8)(x+8)} \cdot \frac{-10(-8+x)}{-9x^2}$

$\frac{2x^3(x+8)(x+4)}{(x-8)(x+8)} \cdot \frac{-10(-8+x)}{-9x^2}$

$\frac{-20x^3(x+4)}{-9x^2}$

$\boxed{\frac{20x(x+4)}{9}}$

⑨ a) Simplify the following polynomial.
b) Write the polynomial in descending order of powers.
c) State the polynomial's degree.

$-6x + 18 - x^2 + 2 + 4x + 10x^2$

a) & b) $\boxed{9x^2 - 2x + 20}$

c) $\boxed{D=2}$

d) Expand: $(5y-4)^2 = (5y-4)(5y-4)$

$$= 25y^2 - 20y - 20y + 16$$

$$= \boxed{25y^2 - 40y + 16}$$

Perform the indicated operations and simplify completely.

⑩ Divide: $\dfrac{18x^8 - 20x^5 + 2x^2}{2x} = \boxed{9x^7 - 10x^4 + x}$

⑪ Expand: $(x+2)(x^2+6x-2)$

$$x^3 + 6x^2 - 2x + 2x^2 + 12x - 4$$

$$\boxed{x^3 + 8x^2 + 10x - 4}$$

⑫ a) Add: $(x^3 - x^2 - x - 1) + (-5x^2 + 7x - 10)$

$$x^3 - x^2 - x - 1 - 5x^2 + 7x - 10$$

$$\boxed{x^3 - 6x^2 + 6x - 11}$$

b) Subtract: $(x+2) - (4x^2 + 2x - 9)$

$$x + 2 - 4x^2 - 2x + 9$$

$$\boxed{-4x^2 - x + 11}$$

⑬ Combine

$$\frac{5}{6x^2+5} + \frac{-7}{6x-6} - \frac{4x-8}{6x^3-6x^2+5x-5}$$

$$\frac{5}{6x^2+5} + \frac{-7}{6(x-1)} - \frac{4x-8}{6x^2(x-1)+5(x-1)}$$

$$\frac{5}{6x^2+5} + \frac{-7}{6(x-1)} - \frac{4x-8}{(x-1)(6x^2+5)}$$

$$\frac{5 \cdot 6(x-1)}{(6x^2+5)\,6(x-1)} + \frac{-7(6x^2+5)}{6(x-1)(6x^2+5)} - \frac{6(4x-8)}{6(x-1)(6x^2+5)}$$

$$\frac{30x-30}{6(x-1)(6x^2+5)} + \frac{-42x^2-35}{6(x-1)(6x^2+5)} - \frac{24x-48}{6(x-1)(6x^2+5)}$$

$$\frac{30x-30-42x^2-35-24x+48}{6(x-1)(6x^2+5)}$$

$$\boxed{\frac{-42x^2+6x-17}{6(x-1)(6x^2+5)}}$$

⑭ Simplify:

$$\frac{36a^2-49b^2}{6a^2+19ab+14b^2} = \frac{(6a-7b)(6a+7b)}{(6a+7b)(a+2b)}$$

$$= \boxed{\frac{6a-7b}{a+2b}}$$

Rationalize the denominators.

(15) $\sqrt{\frac{2}{81}}$

$$\frac{\sqrt{2}}{\sqrt{81}}$$

$$\boxed{\frac{\sqrt{2}}{9}}$$

(16) $\frac{3}{\sqrt{3}}$

$$\frac{3\cdot\sqrt{3}}{\sqrt{3}\cdot\sqrt{3}}$$

$$\frac{3\sqrt{3}}{\sqrt{3\cdot3}}$$

$$\frac{3\sqrt{3}}{\sqrt{9}}$$

$$\frac{3\sqrt{3}}{3}$$

$$\boxed{\sqrt{3}}$$

(17) $\sqrt{\frac{10}{27}}$

$$\frac{\sqrt{10}}{\sqrt{27}}$$

$$\frac{\sqrt{10}}{\sqrt{9\cdot3}}$$

$$\frac{\sqrt{10}}{\sqrt{9}\cdot\sqrt{3}}$$

$$\frac{\sqrt{10}}{3\sqrt{3}}$$

$$\frac{\sqrt{10}\cdot\sqrt{3}}{3\sqrt{3}\cdot\sqrt{3}}$$

$$\frac{\sqrt{10\cdot3}}{3\sqrt{3\cdot3}}$$

$$\frac{\sqrt{30}}{3\sqrt{9}} = \frac{\sqrt{30}}{3\cdot3} = \boxed{\frac{\sqrt{30}}{9}}$$

(18) $$\sqrt{\frac{75}{28}} = \frac{\sqrt{75}}{\sqrt{28}} = \frac{\sqrt{25\cdot3}}{\sqrt{4\cdot7}} = \frac{\sqrt{25}\cdot\sqrt{3}}{\sqrt{4}\cdot\sqrt{7}} = \frac{5\sqrt{3}}{2\sqrt{7}}$$

$$= \frac{5\sqrt{3}\cdot\sqrt{7}}{2\sqrt{7}\cdot\sqrt{7}}$$

$$= \frac{5\sqrt{3\cdot7}}{2\sqrt{7\cdot7}}$$

$$= \frac{5\sqrt{21}}{2\sqrt{49}}$$

$$= \frac{5\sqrt{21}}{2\cdot7}$$

$$= \boxed{\frac{5\sqrt{21}}{14}}$$

(19) Simplify the expressions.

a) $\sqrt{36} = \boxed{6}$

b) $$-8\sqrt{3}(-2\sqrt{6}) = 16\sqrt{3\cdot6} = 16\sqrt{18} = 16\sqrt{9\cdot2}$$

$$= 16\sqrt{9}\cdot\sqrt{2}$$

$$= 16\cdot3\cdot\sqrt{2}$$

$$= \boxed{48\sqrt{2}}$$

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

c) $\sqrt{\frac{49}{100}}$

$\frac{\sqrt{49}}{\sqrt{100}}$

$\boxed{\frac{7}{10}}$

d) $8\sqrt{3} + 2\sqrt{27} - \sqrt{12}$

$8\sqrt{3} + 2\sqrt{9 \cdot 3} - \sqrt{4 \cdot 3}$

$8\sqrt{3} + 2\sqrt{9} \cdot \sqrt{3} - \sqrt{4} \cdot \sqrt{3}$

$8\sqrt{3} + 2 \cdot 3 \cdot \sqrt{3} - 2 \cdot \sqrt{3}$

$8\sqrt{3} + 6\sqrt{3} - 2\sqrt{3}$

$14\sqrt{3} - 2\sqrt{3}$

$\boxed{12\sqrt{3}}$

Simplify.

(20) $\dfrac{\frac{9}{xy}}{-7x^2} = \frac{9}{xy} \div (-7x^2) = \frac{9}{xy} \div \left(\frac{-7x^2}{1}\right)$

$= \frac{9}{xy} \cdot \left(-\frac{1}{7x^2}\right)$

$= \boxed{-\frac{9}{7x^3y}}$

(21) $\dfrac{\frac{3}{x+4} - 2}{\frac{5}{x-4} + \frac{x}{x^2-16}} = \dfrac{\frac{3}{x+4} - \frac{2}{1}}{\frac{5}{x-4} + \frac{x}{(x+4)(x-4)}}$

$= \dfrac{\frac{3}{x+4} - \frac{2(x+4)}{1(x+4)}}{\frac{5(x+4)}{(x-4)(x+4)} + \frac{x}{(x+4)(x-4)}}$

$= \dfrac{\frac{3}{x+4} - \frac{2x+8}{x+4}}{\frac{5x+20}{(x-4)(x+4)} + \frac{x}{(x+4)(x-4)}}$

$= \dfrac{\frac{3-2x-8}{x+4}}{\frac{6x+20}{(x-4)(x+4)}}$

$= \dfrac{\frac{-2x-5}{x+4}}{\frac{6x+20}{(x-4)(x+4)}}$

$= \frac{-2x-5}{x+4} \div \frac{6x+20}{(x-4)(x+4)}$

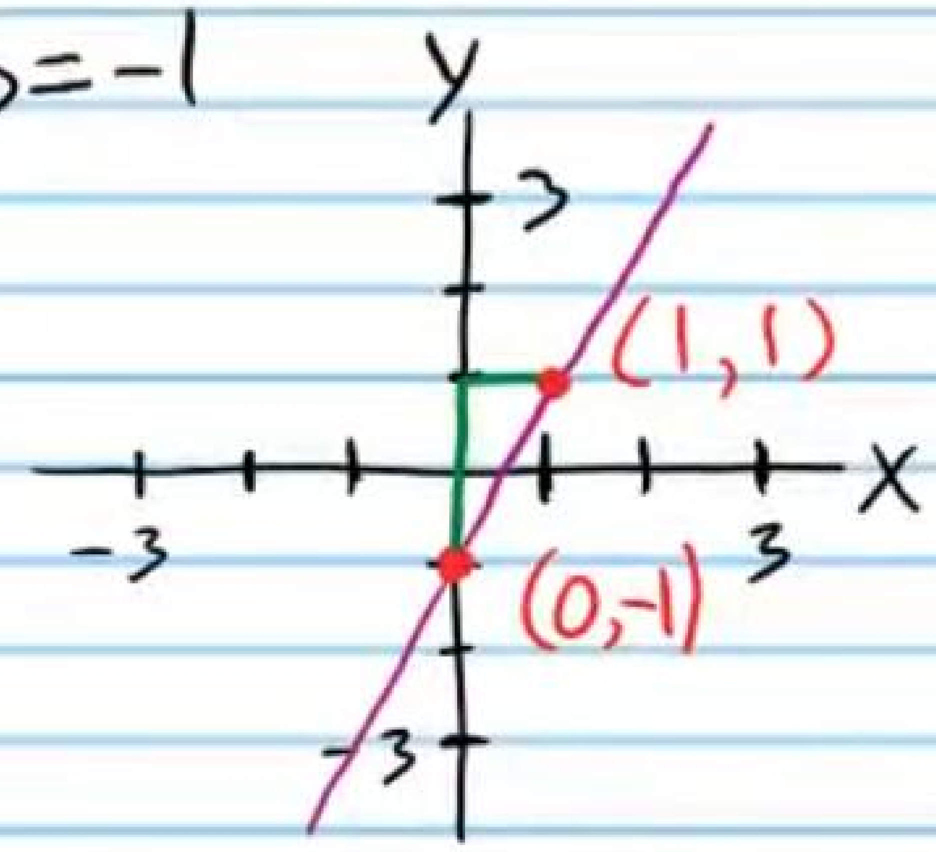
$= \frac{-2x-5}{x+4} \cdot \frac{(x-4)(x+4)}{6x+20}$

$= \boxed{\frac{(-2x-5)(x-4)}{6x+20}}$

For the problems below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,-3)).

(22) Graph the following equation using the slope and y-intercept.

$y = 2x - 1$ $\quad$ $m = \frac{2}{1},\ b = -1$

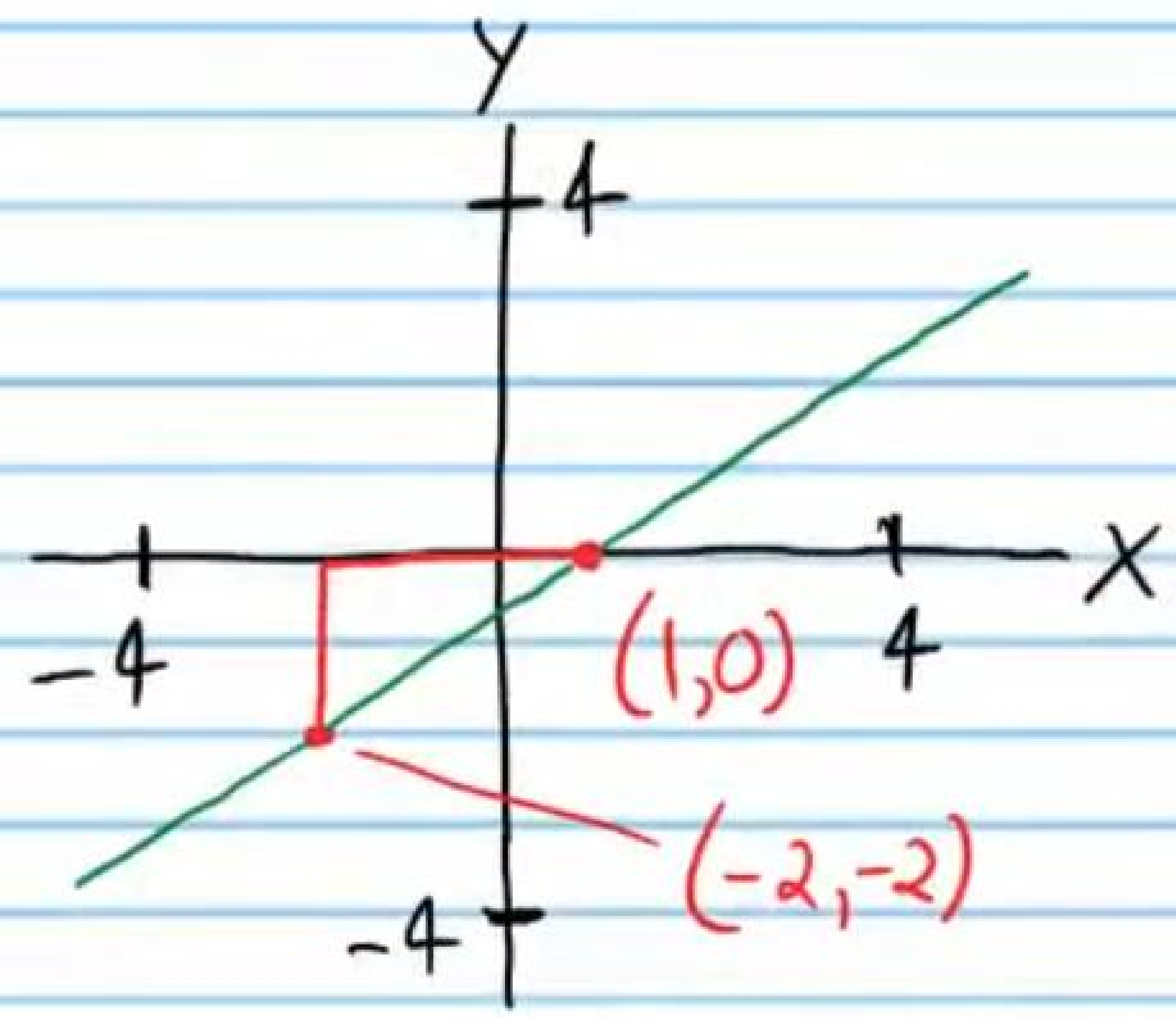


(23) Graph the equation using slope and a point visible in the equation.

$\frac{2}{3}(x+2) = y+2$

$m(x - x_p) = y - y_p$

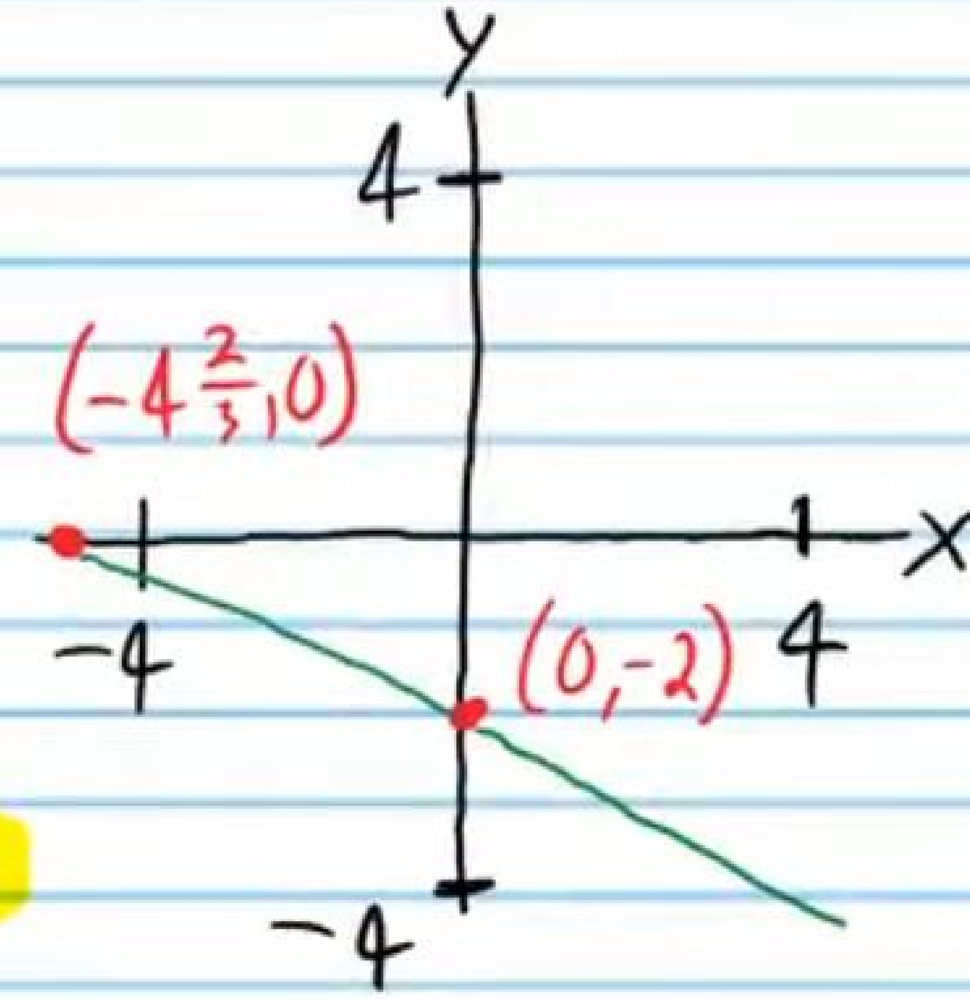
$m = \frac{2}{3},\ (-2,-2)$



(24) Graph the following equation using x and y intercepts.

$-3x - 7y = 14$

Point I	Point II
$x = 0$	$y = 0$
$-3(0) - 7y = 14$	$-3x - 7(0) = 14$
$y = -2$	$x = -\frac{14}{3}$
$(0,-2)$	$x = -4\frac{2}{3}$
	$(-4\frac{2}{3},0)$



(25) Graph the following equation using any method.

$3y = -4x + 1$

Point I	Point II
$x = 0$	$x = 1$
$3y = -4(0) + 1$	$3y = -4(1) + 1$
$3y = 1$	$3y = -4 + 1$
$y = \frac{1}{3}$	$3y = -3$
$(0,\frac{1}{3})$	$y = -1$
	$(1,-1)$